

KRYLOV METHODS FOR THE MINIMUM VARIANCE PORTFOLIO

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Abstract. The long-only minimum variance portfolio is routinely solved by assembling a sample covariance matrix and dispatching it to a general-purpose quadratic programme. Both steps are unnecessary and expensive.

Conjugate gradients applied matrix-free to the returns matrix X solve the problem in $O(nT)$ work per iteration without ever forming the $n \times n$ covariance; a primal-dual active-set loop enforces non-negativity with negligible overhead. The covariance matrix is not a primitive of the problem — it is a computational detour.

Ledoit-Wolf shrinkage, already standard practice for statistical reasons, turns out to be identical to spectral preconditioning of the linear system: the same ridge term that shrinks sample eigenvalues toward their mean also compresses the condition number and cuts CG iteration counts dramatically. Statistical regularisation and numerical preconditioning are the same operation applied to the same matrix — a connection that does not appear to have been made explicit in the existing literature.

Numerical experiments on S&P 500 equity data and synthetic problems of growing size show order-of-magnitude speedups over interior-point solvers. For standard portfolio sizes a direct KKT solve already captures most of the gain; the matrix-free advantage of CG becomes decisive at larger n , where avoiding explicit covariance assembly reduces per-step cost from $O(n^2)$ to $O(Tn)$.

1. Introduction. The minimum variance portfolio allocates capital across n assets so as to minimise return variance subject only to full investment and non-negativity. Since [10] it has been standard practice to estimate a sample covariance matrix $\hat{\Sigma} = X^\top X/T$ from a $T \times n$ return matrix X and pass it to a general-purpose quadratic programming solver. This paper argues that both steps are unnecessary and expensive.

That the minimum-variance solution satisfies $w \propto \Sigma^{-1}\mathbf{1}$ is a classical observation: it follows directly from the KKT stationarity condition $2\Sigma w = \lambda\mathbf{1}$ and was made explicit by [11] in the context of the efficient frontier. The budget constraint is recovered by the single rescaling $w = v/(\mathbf{1}^\top v)$ where $v = \Sigma^{-1}\mathbf{1}$. What has *not* been done is to exploit this structure computationally: the standard pipeline still forms $\Sigma = X^\top X/T$ explicitly before solving.

This paper makes two contributions. First, we apply conjugate gradients (CG) to $\Sigma v = \mathbf{1}$ *matrix-free*: $\Sigma = X^\top X$ is never assembled. Each CG iteration requires only two matrix-vector products with X , reducing per-iteration cost from $O(n^2)$ to $O(nT)$ and eliminating $O(n^2)$ storage. Long-only weights are enforced by a primal-dual active-set loop that calls the CG solver on successively smaller subproblems.

Second, we show that Ledoit-Wolf shrinkage [8] plays a dual role in this framework. Statistically, it reduces estimation error by shrinking sample eigenvalues toward their mean. Numerically, the same modification compresses the eigenvalue spread of the system matrix and directly accelerates CG convergence: on S&P 500 equity data, shrinkage cuts CG iterations from 744 to 82. Statistical regularisation and spectral preconditioning are, in this problem, the same operation applied to the same matrix—a connection that does not appear to have been made explicit in the existing literature.

CG [6] is a cornerstone of large-scale scientific computing [3] but has not been applied to portfolio optimisation in the matrix-free form described here. The closest related work applies GMRES to the minimum variance problem when Σ is singular [1]; that formulation still assembles $X^\top X$ and does not exploit the SPD structure or

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consider preconditioning. General-purpose solvers such as Clarabel [4] handle the long-only constraint natively but require an explicit covariance matrix. ywa Sections 2–5 develop the formulation and solvers; Section 6 reports empirical results on both synthetic and S&P 500 data.

2. Problem Formulation. Let $X \in \mathbb{R}^{T \times n}$ be the demeaned return matrix for n assets over T trading days. The long-only minimum variance problem is:

$$\min_w w^\top \hat{\Sigma} w \quad \text{subject to} \quad \mathbf{1}^\top w = 1, \quad w \geq 0$$

where $\hat{\Sigma} = \frac{1}{T} X^\top X$ is the sample covariance matrix. This is equivalently stated as:

$$\min_w \frac{\|Xw\|^2}{T} = \min_w w^\top \hat{\Sigma} w \quad \text{subject to} \quad \mathbf{1}^\top w = 1, \quad w \geq 0$$

This reframing is key: the objective is now a normalised squared norm of Xw , and we can evaluate $w \mapsto X^\top(Xw)$ using two matrix-vector products, without ever forming $X^\top X$.

Parameter	Type	Description
X	$\mathbb{R}^{T \times n}$	Demeaned return matrix
target	$\mathbb{R}^{n \times n}$	Shrinkage target matrix (optional; default <code>None</code>)
α	$[0, 1]$	Shrinkage intensity; active only when target is provided (default 0)
ρ	$\mathbb{R}_{\geq 0}$	Return-tilt weight (default 0)
μ	$\mathbb{R}^n$	Expected returns (default <code>0</code>)

TABLE 1
Parameters of the Problem specification.

Dual problem.. Including multipliers for both constraints, the full Lagrangian is

$$\mathcal{L}(w, \lambda, \nu) = \frac{\|Xw\|^2}{T} - \lambda(\mathbf{1}^\top w - 1) - \nu^\top w,$$

where $\lambda \in \mathbb{R}$ is the budget multiplier and $\nu \geq \mathbf{0}$ is the vector of non-negativity multipliers. Minimising over w via stationarity $\frac{2}{T} X^\top X w = \lambda \mathbf{1} + \nu$ gives the dual function

$$g(\lambda, \nu) = \inf_w \mathcal{L}(w, \lambda, \nu) = \lambda - \frac{T}{4} (\lambda \mathbf{1} + \nu)^\top (X^\top X)^{-1} (\lambda \mathbf{1} + \nu),$$

defined when X has full column rank; the dual problem is $\max_{\lambda, \nu \geq \mathbf{0}} g(\lambda, \nu)$. The feasible set $\{w : \mathbf{1}^\top w = 1, w \geq \mathbf{0}\}$ is compact and $w = \frac{1}{n} \mathbf{1}$ is strictly feasible, so Slater’s condition holds and strong duality obtains: the primal and dual optima coincide and the duality gap is zero.

At the optimum, complementary slackness ($\nu_i w_i = 0$ for all i) partitions assets into two types. *Active* assets ($w_i > 0$) have $\nu_i = 0$; stationarity then gives $\frac{2}{T} (X^\top X w)_i = \lambda$, so every active asset contributes equally to marginal portfolio variance. *Excluded* assets ($w_i = 0$) have $\nu_i = \frac{2}{T} (X^\top X w)_i - \lambda \geq 0$: their marginal variance is at least λ and they would increase portfolio variance if held at a positive level. The primal–dual loop in Section 3 is a direct implementation of these conditions: the loop terminates precisely when both primal feasibility and dual feasibility hold, which—together with stationarity from each inner solve—is sufficient for global optimality.

Problem variants.. The framework handles a range of standard portfolio construction problems directly. Each is expressed as

$$(1) \quad \min_w \|\tilde{X}w\|^2 - \rho\mu^\top w \quad \text{subject to} \quad \mathbf{1}^\top w = 1, \quad w \geq 0,$$

through choices of $\tilde{X}$, ρ , and μ summarised in Table 2. No change to the algorithm is required; only the data passed to the solver changes.

Portfolio problem	$\tilde{X}$	ρ	μ
Minimum variance	X	0	—
Mean-variance (Markowitz)	X	> 0	expected returns
Minimum tracking error*	X	2	$X^\top(Xb)$
LW-regularised minimum variance	$[\sqrt{(1-\alpha)/T} X; \sqrt{\alpha} R]$	0	—
Factor-model covariance $FF^\top + D$	$[F; D^{1/2}]$	0	—
Regime-blended / multi-period	$[\omega_1 X_1; \dots; \omega_K X_K]$	0	—

TABLE 2

*Portfolio construction problems solved by the same primal–dual loop. b is the benchmark weight vector; F contains factor returns; D is a diagonal specific-risk matrix; R is the $N \times N$ regularisation target matrix (e.g. $\sqrt{\lambda}I$ for classical Ledoit–Wolf shrinkage, where $\bar{\lambda} = \|X\|_F^2/(nT)$ is the mean eigenvalue of $\hat{\Sigma}$). Stacking of $\tilde{X}$ is in the row direction. * $\min_w (w - b)^\top \Sigma (w - b) = \min_w w^\top \Sigma w - 2(\Sigma b)^\top w + \text{const}$, so $\rho\mu = 2\Sigma b = 2X^\top(Xb)$ (two matrix–vector products). Since $\mu = \Sigma b$, the second SPD solve yields $v_2 = b$ exactly; tracking error adds no inner-solve cost.*

What does not fit.. The maximum Sharpe ratio (tangency) portfolio $\arg \max_w \mu^\top w / \sqrt{w^\top \Sigma w}$ subject to $w \geq 0$, is a fractional programme and does not reduce to (1) with a fixed ρ . It can be recovered parametrically by sweeping ρ over the efficient frontier and locating the tangency point, but this requires multiple solves. Problems with additional linear inequality constraints—sector limits, turnover bounds, factor-exposure limits—also lie outside the current template; a general QP solver (e.g. CVXPY with Clarabel) is the appropriate tool for those.

The one structural requirement is that the covariance admit a factorisation $\Sigma = \tilde{X}^\top \tilde{X}$ with $\tilde{X}$ available directly or by cheap row-augmentation, so that the matrix–vector product $v \mapsto \tilde{X}^\top(\tilde{X}v)$ remains $O(T'n)$ per iteration. Nonlinear shrinkage [9], which applies a separate coefficient to each sample eigenvalue, does not admit such a factorisation and requires a different preconditioning strategy.

3. Positive Definite Reduction and Primal–Dual Loop. Ignoring the inequality constraints for now, the Lagrangian for the equality-constrained problem is:

$$\mathcal{L}(w, \lambda) = \frac{\|Xw\|^2}{T} - \lambda(\mathbf{1}^\top w - 1)$$

Setting the derivative to zero gives the stationarity condition $\frac{2}{T}\hat{\Sigma}w = \lambda\mathbf{1} + \rho\mu$, where $\hat{\Sigma} = X^\top X/T$ (or Σ_{LW} with shrinkage). For $\rho = 0$ this reduces to $\frac{2}{T}\hat{\Sigma}w = \lambda\mathbf{1}$, determining the solution up to a scalar.

Positive definite reduction.. When $\rho = 0$ (pure minimum variance), stationarity gives $2\hat{\Sigma}w = \lambda\mathbf{1}$, so w is a scalar multiple of $\hat{\Sigma}^{-1}\mathbf{1}$. Setting $v_1 = \hat{\Sigma}^{-1}\mathbf{1}$ and writing $w = cv_1$, the budget constraint $\mathbf{1}^\top w = 1$ fixes

$$c = \frac{1}{\mathbf{1}^\top v_1}, \quad w = \frac{v_1}{\mathbf{1}^\top v_1}.$$

When $\rho \neq 0$, stationarity gives $2\hat{\Sigma}w = \lambda\mathbf{1} + \rho\mu$, so $w = \frac{\lambda}{2}\hat{\Sigma}^{-1}\mathbf{1} + \frac{\rho}{2}\hat{\Sigma}^{-1}\mu$. Setting $v_1 = \hat{\Sigma}^{-1}\mathbf{1}$ and $v_2 = \hat{\Sigma}^{-1}\mu$ (two SPD solves), the budget constraint determines the multiplier

$$\lambda = \frac{2 - \rho(\mathbf{1}^\top v_2)}{\mathbf{1}^\top v_1},$$

and the weights follow as $w = \frac{\lambda}{2}v_1 + \frac{\rho}{2}v_2$. In both cases the $(n+1) \times (n+1)$ saddle-point system is never needed and λ is recovered analytically rather than as part of the solve.

When $\alpha > 0$ and a regularisation target R is provided, the regularised matrix $\Sigma_{\text{LW}} = \frac{1-\alpha}{T}X^\top X + \alpha R^\top R = (1-\alpha)\hat{\Sigma} + \alpha R^\top R$ is symmetric positive definite and standard CG applies directly to $\Sigma_{\text{LW}}v = \mathbf{1}$. For classical Ledoit-Wolf shrinkage one sets $R = \sqrt{\bar{\lambda}}I$ with $\bar{\lambda} = \|X\|_F^2/(nT)$, giving $\Sigma_{\text{LW}} = (1-\alpha)\hat{\Sigma} + \alpha\bar{\lambda}I$. For completeness, the equivalent $(n+1) \times (n+1)$ saddle-point formulation

$$\begin{pmatrix} 2\Sigma_{\text{LW}} & \mathbf{1} \\ \mathbf{1}^\top & 0 \end{pmatrix} \begin{pmatrix} w \\ \lambda \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ 1 \end{pmatrix}$$

is symmetric but indefinite and requires MINRES [12] rather than CG; it is shown here for completeness.

The long-only constraint $w \geq 0$ is enforced by a **primal-dual iterative procedure**. Each outer step solves the equality-constrained (budget-only) $n_a \times n_a$ positive definite system over the current asset universe. In the *primal step*, any asset whose weight is negative is dropped. Once all active weights are non-negative, a *dual step* checks the KKT gradient condition for every excluded asset: for global optimality, every excluded asset i must satisfy

$$\nabla_i f(w) \geq \lambda, \quad \nabla_i f(w) = 2\left(\frac{1-\alpha}{T}(X^\top X w)_i + \alpha(R^\top R w)_i\right) - \rho\mu_i,$$

where λ is the budget Lagrange multiplier and R is the regularisation target matrix. An excluded asset that fails this condition would decrease portfolio variance if added at a small positive level; it is re-added to the active set and the loop repeats. The loop terminates when both primal and dual feasibility hold simultaneously; combined with stationarity from the KKT solve, this is sufficient for global optimality. The feasible set is non-empty and compact, so the constrained minimum exists.

With Ledoit-Wolf shrinkage, the asset elimination loop converges rapidly in practice. Let $\bar{\lambda} = \|X\|_F^2/(nT)$ denote the average eigenvalue of the sample covariance. The unconstrained minimiser satisfies $w^* \propto \Sigma_{\text{LW}}^{-1}\mathbf{1}$. As $\alpha \rightarrow 1$, $\Sigma_{\text{LW}} \rightarrow \bar{\lambda}I$ and $w^* \rightarrow \frac{1}{n}\mathbf{1}$, the equal-weight portfolio with all components strictly positive. By continuity there exists a threshold $\alpha^* \in (0, 1)$ above which every component of the unconstrained solution is positive; for $\alpha > \alpha^*$ the first solve requires no asset to be dropped. The first-order expansion

$$\Sigma_{\text{LW}}^{-1}\mathbf{1} \approx \frac{1}{\alpha\bar{\lambda}}\mathbf{1} - \frac{1-\alpha}{\alpha^2\bar{\lambda}^2T}X^\top X\mathbf{1},$$

valid for α close to one, shows that α^* is governed by the spectral structure of $X^\top X$: assets with large total covariance exposure are the last to turn positive as α increases. For the LW-optimal α and dense equity returns, the threshold is typically exceeded and the first solve yields few negative components. Across 50 experiments with n from 100 to 1000 and $T = 2n$, the elimination loop required 2–3 outer iterations in all cases; runtime is dominated by the linear solves.

Algorithm 1 accepts any inner solver f ; Section 4 presents four concrete choices: matrix-free CG (the primary method), a direct dense factorisation, non-negative least squares (NNLS), and the CVXPY interior-point reference.

Algorithm 1 Primal-dual outer loop

Require: Inner solver f ; tolerance $\varepsilon = 10^{-6}$; regularisation target R (optional)
Ensure: Weight vector $w \in \mathbb{R}^n$; total inner iteration count k

- 1: $\text{active} \leftarrow \{1, \dots, n\}$; $k \leftarrow 0$
- 2: **loop**
- 3: $(w_a, k_{\text{step}}) \leftarrow f(\text{active})$; $k \leftarrow k + k_{\text{step}}$
- 4: **if** $\exists i \in \text{active}$ with $w_{a,i} < -\varepsilon$ **then**
- 5: $\text{active} \leftarrow \text{active} \setminus \{i : w_{a,i} < -\varepsilon\}$ *(primal step: drop violators)*
- 6: **else**
- 7: $w_i \leftarrow w_{a,i}$ for $i \in \text{active}$; $w_i \leftarrow 0$ for $i \notin \text{active}$
- 8: $g_i \leftarrow 2\left(\frac{1-\alpha}{T}(X^\top X w)_i + \alpha(R^\top R w)_i\right) - \rho\mu_i$
- 9: $\lambda \leftarrow \text{mean}_{j \in \text{active}}(g_j)$
- 10: $\mathcal{V} \leftarrow \{i \notin \text{active} : g_i < \lambda - \varepsilon\}$
- 11: **if** $\mathcal{V} = \emptyset$ **then**
- 12: **break** *(KKT conditions satisfied; globally optimal)*
- 13: **end if**
- 14: $\text{active} \leftarrow \text{active} \cup \mathcal{V}$ *(dual step: re-add violators)*
- 15: **end if**
- 16: **end loop**
- 17: **return** w, k

4. Computational Methods.

4.1. CG Method. The inner solver f for Algorithm 1 is conjugate gradients applied to the $n_a \times n_a$ SPD system derived in Section 3. The action of Σ_{LW} on a vector requires only two matrix-vector products with X_a ; the matrix itself is never formed.

Algorithm 2 CG inner step (inner solver f for Algorithm 1)

Require: Active-asset index set active ; problem data; regularisation target R (optional)
Ensure: Active-asset weights w_a ; CG iteration count k

- 1: **function** CGSOLVE(active)
- 2: Let $X_a \leftarrow X_{\cdot, \text{active}}$, $R_a \leftarrow R[\text{active}, \text{active}]$, $n_a \leftarrow |\text{active}|$
- 3: Build matrix-free operator $\mathcal{S}$ of size $n_a \times n_a$:
 $v \mapsto \frac{1-\alpha}{T} X_a^\top (X_a v) + \alpha R_a^\top (R_a v)$ *(Σ_{LW} applied matrix-free)*
- 4: Solve $\mathcal{S} v = \mathbf{1}$ via CG
- 5: **return** $v / (\mathbf{1}^\top v)$, iteration count
- 6: **end function**

When an asset is dropped in the primal step, $\mathcal{S}$ is rebuilt for the reduced active set and CG restarts from scratch on the smaller system.

Convergence speed.. The classical CG error bound [5] states

$$\frac{\|e_k\|_{\Sigma_{\text{LW}}}}{\|e_0\|_{\Sigma_{\text{LW}}}} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k, \quad \kappa = \kappa(\Sigma_{\text{LW}}),$$

where $\|e\|_{\Sigma} = (e^\top \Sigma e)^{1/2}$ is the energy norm and κ is the spectral condition number introduced in Section 5. Shrinkage compresses κ monotonically as α increases, directly improving the asymptotic convergence rate.

A second, sharper bound follows from the rank structure of Σ_{LW} . Because $X_a \in \mathbb{R}^{T \times n_a}$ has rank at most $\min(T, n_a)$, the matrix $\Sigma_{\text{LW}} = \frac{1-\alpha}{T} X_a^\top X_a + \alpha R_a^\top R_a$ has *at most* $\min(T, n_a) + 1$ distinct eigenvalues: exactly $n_a - \min(T, n_a)$ eigenvalues equal to $\alpha \lambda_{\min}(R_a^\top R_a)$ (the ridge floor) and up to $\min(T, n_a)$ distinct eigenvalues above it. Since CG terminates in exact arithmetic in at most as many steps as there are distinct eigenvalues [5], convergence requires at most $\min(T, n_a) + 1$ iterations. In the underdetermined regime $T < n_a$ this yields a bound of $T + 1$, independent of the active-set size n_a ; adding assets to the portfolio does not increase the iteration count. This regime arises when the history window is shorter than the universe (a common scenario in, e.g., high-frequency or short-window portfolio rebalancing), or within the primal-dual loop once enough assets have been excluded that n_a exceeds T . Shrinkage reinforces the rank-based bound by clustering the $\min(T, n_a)$ non-ridge eigenvalues, causing super-linear acceleration in practice [14]. The iteration counts in Table 3 illustrate the condition-number effect: on the S&P 500 problem ($n = 495$, $T = 1192 > n$) shrinkage reduces CG iterations from 744 to 82, consistent with $\kappa(\Sigma_{\text{LW}})$ decreasing monotonically as α increases (Section 5).

4.2. CVXPY Reference Implementation. Before developing the Krylov solvers we establish a reference solution using CVXPY [2], a domain-specific language for convex optimisation. The minimum variance problem maps directly onto a least-squares objective, which CVXPY dispatches to a general-purpose interior-point solver.

Algorithm 3 CVXPY reference solver

Require: Returns matrix $X \in \mathbb{R}^{T \times n}$; shrinkage α ; return tilt $\rho\mu$; target R

Ensure: Weight vector w ; interior-point iteration count k

- 1: Declare $w \in \mathbb{R}^n$ as a CVXPY variable
 - 2: Minimise $\frac{1-\alpha}{T} \|Xw\|^2 + \alpha \|Rw\|^2 - \rho\mu^\top w$
subject to $\mathbf{1}^\top w = 1$, $w \geq \mathbf{0}$
 - 3: Dispatch to Clarabel interior-point solver [4]
 - 4: **return** w , k
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The formulation is transparent and correct, and CVXPY handles the long-only constraint natively. We use the Clarabel interior-point solver [4], which dispatches the problem as a second-order cone programme. Its cost is substantial: interior-point methods require $O(n^3)$ work per iteration, and the overhead of problem construction and solver communication adds further latency. On the S&P 500 problem ($n = 495$) this amounts to roughly 1.6 seconds (Table 3), making CVXPY unsuitable for production use but ideal as a ground-truth reference.

All three methods in this paper (CVXPY, KKT direct, and CG) solve the same optimisation problem and produce identical portfolio weights up to numerical tolerance. The comparison is purely computational: the question is not which method finds a better portfolio, but which finds the same portfolio fastest.

The speedup figures in Table 3 should be read with this in mind. CVXPY and Clarabel are general-purpose tools: the same code handles additional convex constraints — sector limits, turnover bounds, factor exposure constraints — without modification. The CG method exploits the specific structure of the pure long-only minimum variance problem and would require rework to accommodate constraints beyond the budget and non-negativity requirements. The speedup therefore conflates two effects: genuine algorithmic efficiency from exploiting problem structure, and the overhead of a framework designed for a far broader problem class. For the

standard long-only constraint the full speedup is available; practitioners requiring richer constraint sets face a narrower tradeoff and may find a specialised QP solver a more appropriate baseline.

4.3. Direct Solve (Baseline). For small n , `numpy.linalg.solve` applied to the $n_a \times n_a$ covariance matrix is fast and exact. It requires forming $X_a^\top X_a$ (an $O(n_a^2 T)$ operation), but for $n \ll T$ this is not the bottleneck.

Algorithm 4 Direct solver

Require: Problem data; regularisation target R (optional)

Ensure: Weight vector w ; outer elimination step count s

```

1: function DIRECTSOLVE(active)
2:   Let  $X_a \leftarrow X_{:, \text{active}}$ ,  $R_a \leftarrow R[\text{active}, \text{active}]$ ,  $n_a \leftarrow |\text{active}|$ 
3:   Assemble  $\Sigma_a \leftarrow \frac{1-\alpha}{T} X_a^\top X_a + \alpha R_a^\top R_a$  (SPD,  $n_a \times n_a$ )
4:   Solve  $\Sigma_a v = \mathbf{1}$  by dense factorisation
5:   return  $v / (\mathbf{1}^\top v)$ , 1 (per-step count; no inner iterations)
6: end function
7:  $(w, s) \leftarrow \text{ELIMINATIONLOOP}(\text{DIRECTSOLVE})$  (Algorithm 1)
8: return  $w, s$ 

```

4.4. NNLS Solver. An alternative approach bypasses the primal-dual loop entirely by reformulating the long-only constraint as a non-negative least squares (NNLS) problem. Starting from the objective $\min_w \frac{1-\alpha}{T} \|Xw\|^2 + \alpha \|Rw\|^2$, the budget constraint $\mathbf{1}^\top w = 1$ is incorporated via a large-penalty row, yielding the augmented system:

$$\min_{w \geq 0} \left\| \begin{pmatrix} \sqrt{(1-\alpha)/T} X \\ \sqrt{\alpha} R \\ M \mathbf{1}^\top \end{pmatrix} w - \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \\ M \end{pmatrix} \right\|^2,$$

where $M = \|X\|_F \cdot T$ is a penalty large enough to enforce $\mathbf{1}^\top w \approx 1$ to machine precision. The non-negativity constraint $w \geq 0$ is handled natively by the Lawson-Hanson active-set algorithm [7], which solves NNLS by iterating over the free-variable set rather than forming an explicit KKT system.

This formulation has two practical advantages: it is single-shot (no outer primal-dual loop), and the budget constraint requires no post-processing. However, NNLS does not benefit from the matrix-free structure of CG: the Lawson-Hanson algorithm implicitly forms normal equations $A^\top A$ where A is the augmented matrix, incurring the same $O(n^2 T)$ cost as the direct KKT solve. With Ledoit-Wolf shrinkage, the augmented matrix grows from $T \times n$ to $(T + n) \times n$, making NNLS paradoxically slower with shrinkage than without, an inversion of the pattern seen for KKT and CG. Table 3 confirms this: on the S&P 500 problem NNLS takes 0.038s without shrinkage but 0.086s with it, while CG moves in the opposite direction.

5. Ledoit-Wolf Shrinkage. The sample covariance matrix $\frac{1}{T} X^\top X$ is a poor estimator when n is large relative to T : its smallest eigenvalues are severely downward-biased, and the eigenvalue spread (the ratio of largest to smallest) far exceeds that of the true covariance. Ledoit and Wolf [8] propose replacing the sample covariance with a convex combination of the sample estimate and a scaled identity:

$$\Sigma_{\text{LW}} = (1 - \alpha) \hat{\Sigma} + \alpha \cdot \text{target}, \quad \bar{\lambda} = \frac{\|X\|_F^2}{nT}$$

where $\bar{\lambda}$ is the average eigenvalue of the sample covariance and the classical choice is target $= \bar{\lambda}I$. The implementation uses a general $N \times N$ regularisation target matrix R (satisfying target $= R^\top R$), so that $\Sigma_{\text{LW}} = (1 - \alpha)\hat{\Sigma} + \alpha R^\top R$. Classical Ledoit-Wolf shrinkage is recovered by setting $R = \sqrt{\bar{\lambda}}I$; for i.i.d. $\mathcal{N}(0, 1)$ synthetic returns $\bar{\lambda} \approx 1$ so $R = I$ is equivalent. The shrinkage intensity $\alpha \in (0, 1)$ is chosen analytically to minimise the expected Frobenius distance to the true covariance, requiring no cross-validation. The effect on the spectrum is direct: the target term lifts every eigenvalue, bounding the smallest away from zero and compressing the overall spread. The condition number satisfies:

$$\kappa(\Sigma_{\text{LW}}) = \frac{\lambda_{\max}(\Sigma_{\text{LW}})}{\lambda_{\min}(\Sigma_{\text{LW}})} \leq \frac{(1 - \alpha)\lambda_{\max}(\hat{\Sigma}) + \alpha\lambda_{\max}(R^\top R)}{\alpha\lambda_{\min}(R^\top R)}$$

since $\lambda_{\min}(\Sigma_{\text{LW}}) \geq \alpha\lambda_{\min}(R^\top R)$ and, by Weyl's inequality applied to $\Sigma_{\text{LW}} = (1 - \alpha)\hat{\Sigma} + \alpha R^\top R$, $\lambda_{\max}(\Sigma_{\text{LW}}) \leq (1 - \alpha)\lambda_{\max}(\hat{\Sigma}) + \alpha\lambda_{\max}(R^\top R)$, where $\lambda_{\max}(\hat{\Sigma}) = \lambda_{\max}(X^\top X/T)$ is the largest sample eigenvalue. As $\alpha \rightarrow 1$ the bound approaches $\lambda_{\max}(R^\top R)/\lambda_{\min}(R^\top R)$, which equals 1 for the classical choice $R = \sqrt{\bar{\lambda}}I$, correctly reflecting that full shrinkage yields a scalar multiple of the identity. For fixed data, the condition number decreases monotonically as α increases, with the rate governed by the spectral ratio $\lambda_{\max}(\hat{\Sigma})/\lambda_{\min}(R^\top R)$. For CG applied to $\Sigma_{\text{LW}} v = \mathbf{1}$, the convergence rate is governed by $\kappa(\Sigma_{\text{LW}})$; compressing the spectrum directly reduces iteration counts. Subsequent work by the same authors provides an analytical nonlinear shrinkage formula [9] that applies a different coefficient to each sample eigenvalue, achieving a smaller Frobenius loss when n/T is not negligible. Nonlinear shrinkage cannot be expressed as a simple augmented matrix $\tilde{X}$, so incorporating it into the matrix-free framework would require a separate Krylov-compatible preconditioner rather than a modified return matrix.

In the implementation, shrinkage is applied by augmenting the return matrix rather than forming Σ_{LW} explicitly. Setting $c = 1 - \alpha$, the stacked matrix $\tilde{X} = [\sqrt{c/T} X; \sqrt{\alpha} R]$ satisfies $\tilde{X}^\top \tilde{X} = \frac{c}{T} X^\top X + \alpha R^\top R = c\hat{\Sigma} + \alpha R^\top R = \Sigma_{\text{LW}}$, so any of the solvers above can be applied to $\tilde{X}$ and produce exactly the LW-shrinkage portfolio weights. For $R = \sqrt{\bar{\lambda}}I$ this gives classical Ledoit-Wolf: $\Sigma_{\text{LW}} = (1 - \alpha)\hat{\Sigma} + \alpha\bar{\lambda}I$.

From the perspective of numerical linear algebra, compressing the eigenvalue spectrum of the system matrix is precisely what a preconditioner does. Ledoit-Wolf shrinkage therefore plays a dual role: it reduces statistical estimation error and, as a side effect, accelerates iterative solvers by clustering the eigenvalues. The two motivations are independent but arrive at the same modified matrix.

The proportional identity shrinkage target $\alpha\bar{\lambda}I$ is the simplest choice, but the preconditioning literature suggests sharper alternatives. A factor-model preconditioner, replacing $\alpha\bar{\lambda}I$ with a low-rank-plus-diagonal approximation $BB^\top + D$, would cluster eigenvalues more tightly at the cost of a slightly more expensive application. Incomplete Cholesky factorisation of the shrunk matrix provides a general-purpose option. For rolling-window problems where the covariance changes slowly between periods, a preconditioner built from the previous period's factorisation could be reused with minimal update cost. None of these require forming the covariance matrix explicitly and each is compatible with the matrix-free framework described above.

6. Numerical Results. All experiments were run on an Apple M4 Pro (14-core CPU, 48 GB RAM) under Python 3.12, NumPy 2.4, SciPy 1.17, CVXPY 1.8.2, and Clarabel 0.11.1. The iterative solvers use SciPy's default relative residual tolerance of 10^{-5} , which was reached in all reported runs; reported times are the minimum of

three repeated solves and cover only the solve itself — construction of the shrinkage target $\hat{\sigma}^2 I$ is excluded, as it requires a single pass over X and is negligible relative to any solver. All implementations are available in the companion package [13].

S&P 500 benchmark. The dataset consists of daily percentage returns for 495 S&P 500 constituents over the period July 2021–May 2026 (1192 trading days, $T/n \approx 2.4$), sourced from Yahoo Finance via the `yfinance` Python package. Tickers with more than 5% missing observations are excluded; short gaps are forward-filled. The return matrix X is formed by demeaning the returns column-wise.

Spectrum of $\hat{\Sigma} = \frac{1}{T} X^\top X$. The eigenvalue structure of the sample covariance is qualitatively different from that of a random Gaussian matrix and is the reason the S&P 500 problem is a meaningful test case for the preconditioning argument. For a random $T \times n$ matrix with i.i.d. $\mathcal{N}(0, \sigma^2)$ entries, the Marchenko-Pastur law predicts that the bulk of the spectrum lies in the interval $[\sigma^2(1 - \sqrt{n/T})^2, \sigma^2(1 + \sqrt{n/T})^2]$, giving a mild condition number that grows modestly with n/T . The equity spectrum is nothing like this.

The largest eigenvalue is $\lambda_1 \approx 0.067$, the market factor, which alone accounts for $\approx 30\%$ of total portfolio variance. To apply the Marchenko-Pastur law we estimate σ^2 as the average squared entry of X ,

$$\hat{\sigma}^2 = \frac{\|X\|_F^2}{nT} \approx 0.000457,$$

the natural entry-level variance of a daily percentage return. The MP upper edge $\hat{\sigma}^2(1 + \sqrt{n/T})^2 \approx 0.00123$ then identifies 22 eigenvalues as statistically significant factors; the remaining 473 are consistent with noise yet still span a wide range down to $\lambda_{\min} \approx 6 \times 10^{-7}$. The resulting condition number is

$$\kappa(\hat{\Sigma}) = \frac{\lambda_{\max}}{\lambda_{\min}} \approx 109,000,$$

far larger than the $O(10)$ – $O(100)$ range typical of random matrices at this aspect ratio. The first ten eigenvalues account for half the total variance, while the bottom half of the spectrum contributes less than 3%. This extreme imbalance—a handful of dominant factors and a long tail of near-zero noise eigenvalues—is precisely what makes CG slow without preconditioning (744 iterations) and what makes shrinkage decisive: lifting $\lambda_{\min}$ by even a moderate α collapses the condition number and cuts the iteration count to 82.

Table 3 reports timings and iteration counts for all solvers on this dataset.

The real-data results reveal two effects. First, without shrinkage the equity system is hard to solve iteratively: CG requires 744 iterations and is slower than KKT direct (0.0120 s vs. 0.0056 s). The strong factor structure of equity returns concentrates variance along a few eigendirections, inflating the condition number and offsetting the matrix-free advantage of CG.

Second, Ledoit-Wolf shrinkage benefits both iterative and direct solvers: it cuts CG iterations from 744 to 82 and reduces active-set outer steps from 22 to 6, since a lower condition number means fewer assets end up with negative unconstrained weights and the primal-dual loop converges faster. At $n = 495$, KKT direct (0.0062 s, $399\times$ over CVXPY) and CG (0.0042 s, $594\times$) are both well under a millisecond and within a factor of 1.5 of each other. For portfolios of this size KKT direct is the practical recommendation: it requires no convergence tolerance and is faster without shrinkage. CG’s advantage lies at larger n , where avoiding explicit assembly of $\tilde{X}^\top \tilde{X}$ reduces

Method	Time (s)	Iterations	Speedup
<i>Without Ledoit-Wolf shrinkage</i>			
cvxpy (Clarabel)	1.778	11	1.0x
Clarabel (direct API)	0.071	11	25.1x
KKT direct	0.0056	22	315.3x
CG (SPD)	0.0120	744	148.1x
NNLS (Lawson–Hanson)	0.0368	–	48.4x
<i>With Ledoit-Wolf shrinkage ($\alpha = 0.5$)</i>			
cvxpy (Clarabel)	2.482	16	1.0x
Clarabel (direct API)	0.092	16	27.0x
KKT direct	0.0062	6	399.0x
CG (SPD)	0.0042	82	594.4x
NNLS (Lawson–Hanson)	0.1113	–	22.3x

TABLE 3

S&P 500 universe: 495 assets, 1192 trading days (Jul 2021–May 2026). Speedup relative to cvxpy within each panel. “Clarabel (direct API)” calls the same interior-point solver without CVXPY’s overhead; > 95% of cvxpy’s time is Python interface work. NNLS is slower with shrinkage because the augmented matrix grows from $T \times n$ to $(T + n) \times n$.

per-step cost from $O(n^2)$ to $O(Tn)$ (Figure 1). The “Clarabel (direct API)” row confirms the speedup is not merely framework overhead: bypassing CVXPY reduces Clarabel’s time from 2.5 s to 0.092 s, yet KKT direct (0.006 s) is still 15× faster. These results are specific to this five-year window; iteration counts depend on the factor structure of equity returns and may differ in other market regimes.

Scaling with problem size. Figure 1 uses synthetic i.i.d. Gaussian data to isolate the computational scaling of each solver across a range of problem sizes and shrinkage levels; the spectral properties of the data are irrelevant here since the point is to measure cost growth, not iteration counts. The left panel shows wall-clock time as a function of n (with $T = 2n$ and LW shrinkage throughout). The KKT direct solve scales as $O(n^3)$, dominated by the $n \times n$ linear system. Each Krylov iteration requires $O(nT) = O(n^2)$ work for the two matrix-vector products with X ; with the spectrum compressed by shrinkage the iteration count grows slowly with n , giving an empirical scaling close to $O(n^2)$ and a gap that widens visibly beyond $n \approx 300$.

The right panel uses the rank-deficient setting $T = 250 < n = 500$, in which $X^\top X$ has at most T non-zero eigenvalues and the unshrunk sample covariance is singular. This regime is practically relevant: asset universes with more instruments than trading days arise when securities with limited common history are pooled, or when a large cross-section is combined with a short lookback window. Without shrinkage the KKT system is singular and the iterative solvers do not converge; every point in the panel has $\alpha > 0$. The preconditioning effect of shrinkage is strongest precisely here: iteration counts fall steeply as α increases from zero, reaching the practical operating point at moderate α values well below unity.

7. Conclusion. The long-only minimum variance portfolio problem, formulated directly in terms of the returns matrix, reduces to solving the $n \times n$ symmetric positive definite system $\Sigma_{\text{LW}} v = \mathbf{1}$ via conjugate gradients, followed by a single rescaling $w = v/(\mathbf{1}^\top v)$. The covariance matrix is never formed; all operations are matrix-vector products with X .

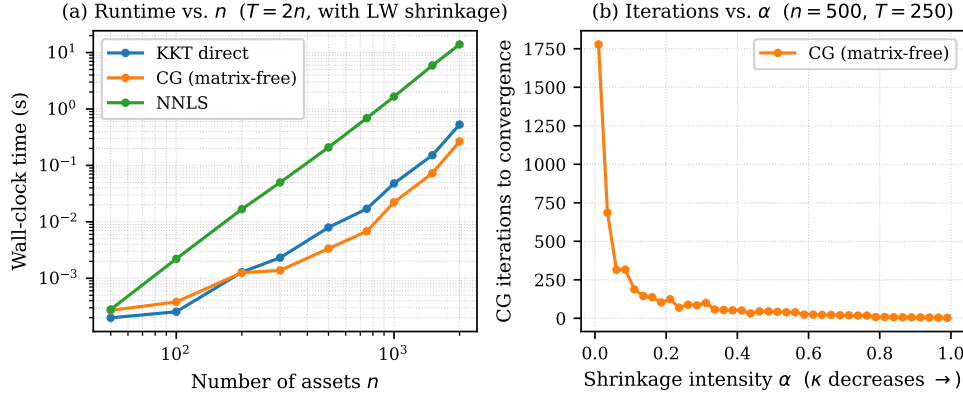


FIG. 1. Left: wall-clock time vs. number of assets n (log-log scale, $T = 2n$, Ledoit-Wolf shrinkage applied throughout) for KKT direct, CG (matrix-free), and NNLS (Lawson-Hanson). Right: CG iteration counts vs. shrinkage intensity α at fixed $n = 500$, $T = 250$ (rank-deficient baseline). Larger α compresses the eigenvalue spectrum and reduces the condition number $\kappa(\Sigma_{\text{LW}})$, directly reducing CG iterations.

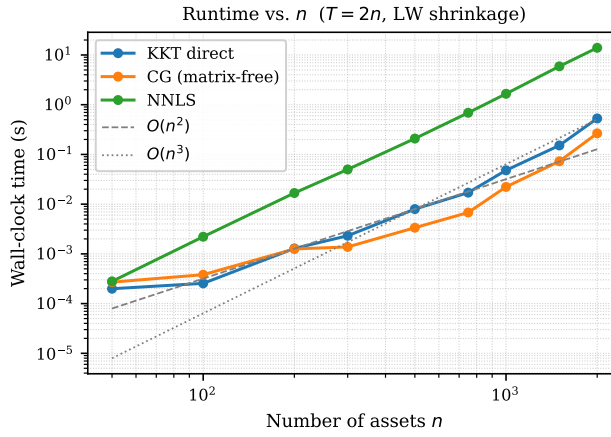


FIG. 2. Wall-clock time vs. number of assets n (log-log scale, $T = 2n$, Ledoit-Wolf shrinkage applied throughout) for KKT direct, CG (matrix-free), and NNLS (Lawson-Hanson). KKT direct scales as $O(n^3)$; matrix-free CG scales empirically as $O(n^2)$ due to the iteration count growing slowly with the compressed spectrum; NNLS also scales as $O(n^3)$ (Lawson-Hanson normal equations). Reference slopes are anchored to the KKT curve at $n = 500$.

The empirical results on 495 S&P 500 constituents over five years confirm the practical value of this approach. For portfolios up to $n \approx 1000$, KKT direct is already 300–400 $\times$ faster than CVXPY across both shrinkage regimes and requires no convergence tolerance: it is the practical recommendation. Without shrinkage, the matrix-free advantage of CG is offset by high iteration counts (744 on S&P 500 equity data) driven by the strong factor structure of equity returns. Ledoit-Wolf shrinkage [8] benefits both solvers: it compresses the eigenvalue spectrum, cuts CG iterations to 82, and reduces active-set outer steps from 22 to 6, since a lower condition number shrinks the unconstrained minimum-variance portfolio toward equal weights and fewer assets receive negative weights in the primal step. At $n = 495$ CG (0.0042 s) edges

KKT direct (0.0062s) by a factor of 1.5. CG’s advantage becomes decisive at large n : the scaling experiments show CG is roughly $2\times$ faster than KKT at $n = 2000$, a gap that widens further as n grows.

The shrinkage estimator of Ledoit and Wolf and the preconditioning theory of Golub and Van Loan [3] address different problems but act on the same object: the eigenvalue spectrum of the system matrix. This connection, which the two literatures do not usually make explicit, points toward richer preconditioners (factor-model or low-rank structured) as a productive direction for future work.

The SPD structure extends naturally beyond the minimum variance setting. Adding a return term $\rho\mu^\top w$ to the objective requires solving two systems ($\Sigma_{\text{LW}} v_1 = \mathbf{1}$ and $\Sigma_{\text{LW}} v_2 = \mu$) and recovering the budget multiplier analytically; the full Markowitz mean-variance problem is handled in a companion paper.

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